OOS Number: OOS- 262333

Phase I

Section A – Test Details					
Initiator:	Cuong Du (Written by Qiyue (	**Test Date:**	05-Oct-2026	**Date Initiated:**	12-Oct-2026
Date of Incident:	12-Oct-2026	**Test Name:**	Celsis Sterility Test	**Sample ID:**	ETX-261002-0288
Name of Analyst who Performed the Test:	Prepping Analyst: Mukyung Jang (MJ) Processing Analyst: Alex Saravia (ES) Aliquoting Analyst: Cuong Du (CCD) Reading Analyst: Cuong Du (CCD)	**Sample / Active Name**	TESAMORELIN 8 MG/ML (3 ML) Injection	**Lot #:**	LG342011076
		Dosage Form	Injection		
Description of Incident:	On 12 Oct 2026, sample ETX-261002-0288 was found to be positive for growth after Celsis Sterility Testing.	☑ Client Care notified of OOS ☑ Sterility and Endotoxin test related OOS; Client must be notified within 24 hours of OOS			
	SOP / Test Method # MICRO-SOP-44	Effective Date: 03-Jan-2025	SOP / Test Method Rev: 01	Limits/Specification: Pass/Fail	
Manager notified	Manager/designee name: Kathan Parikh	Date: 12-Oct-2026	Quality Assurance Department notified:	Yes ☑ No ☐	

Section B – Investigation and Root Cause Analysis			
Personnel Discrepancy		**Comments**	
Analyst interviewed?	☑Yes ☐ No ☐ N/A	Yes, analysts Mukyung Jang, Alex Saravia, and Cuong Du were comprehensively interviewed.	
Correct samples analyzed?	☑Yes ☐ No ☐ N/A	Yes, Sample ID ETX-261002-0288	
Correct SOP / Test Method used/followed?	☑Yes ☐ No ☐ N/A	Yes, as per MICRO-SOP-44	
Correct sample collection/technique?	☑Yes ☐ No ☐ N/A	Yes, as per MICRO-SOP-44	
Are the Test Result Records Attached & Calculations Verified and Correct?	☑Yes ☐ No ☐ N/A	Yes, Information is available in EagleTrax under ETX-261002-0288	
Is the analyst qualified to perform the test?	☑Yes ☐ No ☐ N/A	Yes, all analysts are trained and qualified by the quality department to perform the test	
Is formula worksheet provided and verified?	☐Yes ☐ No ☑ N/A	Not Applicable	
Does the processing method/method set conform to the required specification?	☐Yes ☐ No ☑ N/A	Not Applicable	
Was the sample stored appropriately?	☑Yes ☐ No ☐ N/A	Yes, the sample was stored refrigerated as per client's instructions	
Material Discrepancy		**Lot Number**	**Expiry Date**
Is the integrity of the sample(s) questionable?	☐Yes ☑ No ☐N/A		
Was the sample transported appropriately?	☑Yes ☐ No ☐N/A		
Are the other test results from this run under investigation?	☐Yes ☑ No ☐N/A		

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Appropriate/correct glassware used?	☐Yes ☐ No ☑N/A		
Are reagents stored in proper condition and not expired?	☑Yes ☐ No ☐N/A	Materials and Reagents TSB (300mL): Lot: 08132026-1 FTM (300mL): Lot: 08182026-6 Tween (Polysorbate) 80: Lot: 0000743947 Celsis AmpiScreen Reagents: 022601-1400 Celsis Daily Wash & Rinse Kit: 012503-1142N Celsis ATP Positive Control: 022601-1483 TSA and Surface Plates: Please refer to the Environmental Logs uploaded under the submission on EagleTrax Monthly Cleaning:	Materials and Reagents TSB: Exp: 11Nov2026 FTM: Exp: 16Nov2026 Tween 80: Exp: 31Jan2030 Celsis AmpiScreen Reagents: Jan 2027 Celsis Daily Wash & Rinse Kit: Jan 2027 Celsis ATP Positive Control: Jan 2027 TSA and Surface Plates: Please refer to the Environmental Logs uploaded under the submission on EagleTrax Monthly Cleaning: HaloMist: Exp: 06Jun28 H2O2 strips: Exp: 20Jun28
Standard Used	Celsis ATP Positive Control	022601-1483	31 Jan 2027
Solvent Used	Not Applicable	Not Applicable	Not Applicable
Is the standard the source of the root cause?	☐Yes ☐ No ☑N/A		
Has the analyst been instructed to keep the original test solution for possible retesting?	☐Yes ☐ No ☑N/A		
Equipment Discrepancy		**Equipment ID**	**Calibration Due Date**
Was the instrument set up correctly?	☑Yes ☐ No ☐ N/A	E002222	Dec 2026
Is the clean room facility certified?	☑Yes ☐ No ☐ N/A	E001736 (CR 114), BSC E001316	Dec 2026, Jun 2027 Dec 2026, Jun 2027
System Performance Check (if applicable)?	☑Yes ☐ No ☐ N/A	E002222	Dec 2026
System Suitability RSD < 2% (PDA)?	☐Yes ☐ No ☑ N/A	Not Applicable	Not Applicable
System Suitability RSD < 10% (QDA, CAD)?	☐Yes ☐ No ☑ N/A	Not Applicable	Not Applicable
Do the environmental monitoring results conform to required standards?	☑Yes ☐ No ☐ N/A	See Phase I Summary	See Phase I Summary
Do the controls (i.e., positive, negative, etc.) conform to the required standards?	☑Yes ☐ No ☐ N/A	See Phase I Summary	See Phase I Summary
Is equipment/instrument functional and calibrated/qualified/certified/PM?	☑Yes ☐ No ☐ N/A	Incubator E001356 (Sensor E001450) Incubator E001357 (Sensor E001449) Incubator E001034 (Sensor E001501) Incubator E001031 (Sensor E001505) ISO 5 BSC E001316 ISO 5 BSC E001798	Jan 2027 Feb 2027 Jan 2027 Feb 2027 Aug 2027 Feb 2027 Aug 2027 Feb 2027 Jun 2027 Jun 2027

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Are chromatogram/spectra/raw data correct?	☐ Yes ☐ No ☑ N/A	Not Applicable	Not Applicable

Section C – Evaluation of Laboratory Testing/Hypothesis Testing (if applicable)

- ☑ N/A
- ☐ Re-injection of sample vial
- ☐ Re-injection of original standard vial
- ☐ Re-vialing of original working standard solution
- ☐ Re-vialing of original working sample solution
- ☐ Re-dilution of a working sample solution from the original stock
- ☐ Re-dilution of a working standard solution from the original stock standard
- ☐ Other: N/A QYC 22Oct26

Section D – Phase I Summary

Phase I Summary:

All analysts involved in the preparation (Mukyung Jang), processing (Alex Saravia), aliquoting (Cuong Du), and reading of the samples (Cuong Du) were interviewed comprehensively. Their responses are documented throughout this report. This investigation was performed as per MICRO-SOP-53 Sterility Test Out-of-Specification (OOS) Investigation Procedure.

Upon arrival on 02Oct26, the sample was stored refrigerated in accordance with the client instructions. Both sample prepping analyst Mukyung Jang and processing analyst (Alex Saravia) inspected the sample during preparation and processing. No leaks or visible turbidity were observed, and no evidence of compromised container integrity was identified.

All reagents and supplies mentioned in the material section above were stored according to supplier recommendations, and their integrity was visually verified before utilization. Moreover, all reagents and supplies had valid expiration dates. The functionality of all equipment was confirmed by reviewing data generated by our comprehensive in-house continuous monitoring system.

The cleanroom suite used for processing procedures (CR114) comprises three interconnected sections: the innermost ISO 7 cleanroom (114B), which connects to the middle ISO 7 buffer room (114A), and then to the outermost ISO 8 anteroom (114). A positive air pressure system is maintained throughout the suite to ensure controlled, unidirectional airflow from 114B through 114A and into 114.

The innermost ISO 7 cleanroom 114B and its ISO 5 BSC E001316 were thoroughly cleaned and prepared before initiating testing by analyst Alex Saravia as per MICRO-SOP-2 (Environmental Monitoring of the Clean Room Facility) and MICRO-SOP-9 (Cleaning and Disinfecting Procedure for Microbiology). For Celsis aliquoting, intermediate ISO 7 buffer room 114A and its ISO 5 BSC E001798 were thoroughly cleaned and prepared by analyst (Cuong Du) before initiating testing as per MICRO-SOP-44 and MICRO-SOP-9. Both BSCs, E001316 in 114B and E001798 in 114A, were certified and approved by the Engineering and Quality Assurance teams prior to use.

On 05 Oct 2026, the sample vials for ETX-261002-0288 were received from the Sample Submissions team and brought into the Sterile Microbiology lab. Upon arrival, each vial was sprayed with an acidified bleach disinfectant, placed into pre-disinfected bins, and allowed a 10-minute contact time. Secondary disinfection was performed in the ISO 8 anteroom (114), where the vials were again treated with acidified bleach with a 10-minute contact time before transfer into cleanroom 114B. Inside 114B, the processing analyst (Alex Saravia) performed a final disinfection step with a 10-minute contact time. Once fully disinfected, the vials were transferred into the ISO 5 BSC E001316, placed on the disinfected work surface, aseptically opened, and tested in accordance with MICRO-SOP-44. Direct inoculation was performed by adding 15 mL of sample per media type (divided across ten 3 mL vials), using one 300 mL FTM jar and one 300 mL TSB jar. In accordance with method suitability study ETX-250826-0390, 3 mL of sterile Tween 80 was added into each 300 mL media bottle. Accordingly, a matching negative control jar with the same modifications was made. Following testing, the media bottles were transferred into incubators E001356 and E001357 to initiate incubation.

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Phase I Summary (continued):

Upon completion of incubation on 12 Oct 2026, the media bottles for ETX-261002-0288 were disinfected and transferred to the middle ISO 7 buffer room (114A) for the aliquoting step per MICRO-SOP-44 (Celsis Sterility Testing). In cleanroom 114A, the media bottles were disinfected one more time before transfer into the ISO 5 BSC E001798 located in cleanroom 114A. In ISO 5 BSC E001798, the sample was aliquoted into assay cuvettes by aliquoting analyst (Cuong Du). Following aliquoting, Celsis Sterility Reading was performed on instrument Celsis Advance 2 E002222 under workload 101226-2222-5.wkl in accordance with MICRO-SOP-44 by reading analyst (Cuong Du).

Following the reading, sample ETX-261002-0288 was found to yield a positive reading in the 300 mL FTM media jar. The confirmed average Relative Luminescence Units (RLU) from the duplicate reading tubes, originating from the positive FTM sample bottle, yielded 3,647,568 RLU (duplicate tube readings of 3,608,783 RLU and 3,686,352 RLU, %CV 1.5%), which exceeded the positive cutoff of 2,596.5 RLU, where the FTM negative control was 866 RLU. The 300 mL TSB sample bottle tested negative at 776 RLU (cutoff 2,326.5 RLU). The %CV from the duplicate reading tubes for the positive FTM bottle was well within the specification (< 30%). Additionally, all Daily Controls, including the Instrument Blank (29 RLU, cutoff 100), Reagent Blank (103 RLU, cutoff 1,000), and Celsis ATP Positive Control (99,973 RLU, cutoff 50,000), were within the defined specifications, each with a %CV below 30%.

Following the OOS result, the positive FTM media jar for ETX-261002-0288 was submitted for subculture and identification to isolate viable organisms, and to determine the potential origin of the contamination. Subculture of the positive FTM bottle under tracking ID ETX-261012-0654 yielded viable microbial growth, and confirmatory Oxford Nanopore sequencing identified the isolate as Bacillus paralicheniformis (Gram-positive spore-forming rod, 85.73% match). The positive reading and identification for the sample are documented in Table 1.

All media used were within their expiration dates. Negative controls were prepared and incubated alongside the samples under the same conditions, and all negative controls passed. Monthly cleaning and disinfection of Cleanroom Suite 114 (including 114, 114A, 114B, and the BSCs E001316 and E001798) were performed on 27Sep26 by Sasha Allen (SAA), as per MICRO-SOP-9 (Cleaning and Disinfecting Procedure for Microbiology). Decontamination was performed using HaloMist (Lot 2606A, Exp 06Jun28), and all chemical indicator strips (Lot 260620, Exp 20Jun28) passed, verifying that environmental sanitization remained validated and in control prior to testing. Attached Table 2 and Table 3 present the environmental monitoring results spanning the testing period. All environmental monitoring plates were incubated for no less than 48 hours at 30 to 35°C and no less than an additional 5 days at 20 to 25°C as per MICRO-SOP-2 (Environmental Monitoring of the Clean-room Facility). Table 2 pertains to Environmental Monitoring performed during Celsis Sterility Processing, and Table 3 pertains to Environmental Monitoring performed during Celsis Sterility Aliquoting.

A review of the Environmental Monitoring (EM) results for the relevant Celsis Sterility processing period showed no microbial growth on personnel monitoring plates, settling plates, or ISO 5 BSC E001316 surface sampling plates on the date of testing (05Oct26), the preceding sampling date (02Oct26), or the subsequent sampling date (06Oct26). Weekly active air and surface sampling of the cleanroom suite associated with processing was performed on 05Oct26. No microbial recovery (No Growth) was observed from weekly active air or surface sampling of the cleanroom suite.

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Phase I Summary (continued):

After reviewing the Environmental Monitoring (EM) results for the relevant Celsis Sterility aliquoting period, no microbial growth was observed on personnel monitoring plates, ISO 5 BSC E001798 surface sampling plates, or settling plates on the date of aliquoting (12Oct26), the preceding sampling date (09Oct26), or the subsequent sampling date (13Oct26). Weekly surface sampling of the cleanroom suite performed on 12Oct26 showed no growth. Weekly active air sampling of the cleanroom suite performed on 12Oct26 showed recovery of 6 CFUs (ETX-261021-0578) in the outermost ISO 8 area (room 114), with zero recovery in the ISO 7 buffer or cleanrooms. Confirmatory Oxford Nanopore sequencing identified the recovered isolates as Moraxella osloensis, Sporisorium graminicola, Staphylococcus haemolyticus, Staphylococcus hominis, Kocuria indica, Ustilago maydis, and Kocuria rhizophila/ sp. BT304.

Evaluation of Environmental Monitoring Results:

It is important to note that, for both processing and aliquoting activities, the samples are processed within validated ISO 5 BSCs (E001316 and E001798 respectively). Although microbial recovery was observed from weekly active air monitoring in the outermost ISO 8 room 114 background environment in cleanroom suite (CR114), these recoveries were not obtained from the ISO 5 processing environment or the ISO 7 cleanroom environment.

The absence of microbial recovery from both the processing and the aliquoting analyst glove plates, ISO 5 BSC surface monitoring plates, and ISO 5 settling plates indicates that no identifiable contamination transfer pathway existed between the monitored background cleanroom environment and the ISO 5 critical processing areas. Samples were also prepared and transferred into disinfected, lidded containers and transported on carts within the controlled cleanroom environment. Additionally, none of the 7 environmental isolates recovered from the ISO 8 anteroom belong to the genus Bacillus or exhibit spore-forming rod morphology, demonstrating that they are distinctly different from the Bacillus paralicheniformis recovered from the positive product sample. Therefore, while the active air results demonstrate microbial recovery in the surrounding ISO 8 area, there is no evidence that the recovery originated from, or affected, the ISO 5 BSC environment where the test sample was handled and processed.

All analysts confirmed full compliance with cleaning procedures as outlined in MICRO-SOP-9 (Cleaning and Disinfecting Procedure for Microbiology) and MICRO-SOP-44 (Celsis Sterility Testing). A review of the available data confirms that the cleanroom and equipment conditions remained within acceptable parameters. No deviations or obvious signs of laboratory error during processing and aliquoting were noted from analysts.

No other sample processed by analyst Alex Saravia on 05 Oct 2026 and aliquoted by analyst Cuong Du on 12 Oct 2026, failed Celsis Sterility testing that day. To assess the potential for sample-to-sample contamination contributing to the positive results, a comprehensive review was conducted of all samples processed in the same batch. All other samples processed by the same analyst on the day of testing were found to test negative for microbial growth. These findings suggest that cross-contamination between samples is unlikely.

Analyzing a 6-month sample history for Optimal Balance Pharmacy indicates, this specific analyte "TESAMORELIN 8 MG/ML (3 ML) Injection" has had no prior failures using the Celsis Sterility testing during this period.

Based on the laboratory investigation, no assignable laboratory cause was identified involving the analyst, instrument, reagents, supplies, or monitored laboratory environment. The initial Celsis OOS result therefore remains valid in accordance with the applicable laboratory OOS procedure.

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Has root cause been identified as a possible laboratory error?
Note: Root cause must be conclusive, verifiable, and well-documented.

☐ Yes

☐ If retesting is ***required***, complete Phase II of the OOS (initiate Form 3.100.019.F02).

☑ No, the OOS result is valid.

Reviewer comment(s):

Not Applicable

Prepared by (*Print*)**:**	Signature/Date:
Qiyue Chen	
Lab Manager/Designee (*Print*)**:**	Signature/Date:
Robin Seymour	

Section E – Quality Disposition

Can investigation be closed?

☐ Yes

☐ Original result is valid.

☐ No, initiate Form 3.100.019.F02.

Approved by QA Representative (*Print*)**:**	Signature/Date:

Table 1: Information for ETX-261002-0288 under investigation

Processing Analyst	Aliquoting Analyst	Sample ID	Related Microbial ID	Media with microbial growth	Microbial ID
Alex Saravia	Cuong Du	ETX-261002-0288	ETX-261012-0654	1 x 300mL FTM	*Bacillus paralicheniformis*

Table 2: Environmental Monitoring from Processing Performed on 05Oct26

Environmental Monitoring (EM) Sampling Site	Frequency	Date (DDMMMYY)	Analyst (Initials)	Day /Week(s)	Observation*	Environmental Monitoring Plate ETX ID	Microbial ID	Notes
Personnel EM Bracketing								
Personnel (Left Touch and Right Touch)	Daily	02Oct26	ES	Date Before Testing	No Growth	Not Applicable	None	None
Personnel (Left Touch and Right Touch)	Daily	05Oct26	ES	Date of Testing	No Growth	Not Applicable	None	None
Personnel (Left Touch and Right Touch)	Daily	06Oct26	ES	Date After Testing	No Growth	Not Applicable	None	None
Biological Safety Cabinet EM Bracketing Biological Safety Cabinet (BSC)								
Surface Sampling of ISO 5 (4 locations)	Daily	02Oct26	ALA	Date Before Testing	No Growth	Not Applicable	None	None
Surface Sampling of ISO 5 (4 locations)	Daily	05Oct26	ES	Date of Testing	No Growth	Not Applicable	None	None
Surface Sampling of ISO 5 (4 locations)	Daily	06Oct26	ES	Date After Testing	No Growth	Not Applicable	None	None
Settling Sampling of ISO 5 (2 locations)	Daily	02Oct26	ALA	Date Before Testing	No Growth	Not Applicable	None	None
Settling Sampling of ISO 5 (2 locations)	Daily	05Oct26	ES	Date of Testing	No Growth	Not Applicable	None	None
Settling Sampling of ISO 5 (2 locations)	Daily	06Oct26	ES	Date After Testing	No Growth	Not Applicable	None	None
Weekly Active Air Sampling Bracketing								
Active Air Sampling of Cleanrooms	Weekly	05Oct26	SMO	Week of Testing	No Growth	Not Applicable	None	None
Surface Sampling of Anteroom and Cleanroom Bracketing								
Surface Sampling of Cleanrooms	Weekly	05Oct26	SMO	Week of Testing	No Growth	Not Applicable	None	None

*Observation is based on CFU counts

Table 3: Environmental Monitoring from Aliquoting Performed on 12Oct26

Environmental Monitoring (EM) Sampling Site	Frequency	Date (DDMMMYY)	Analyst (Initials)	Day /Week(s)	Observation*	Environmental Monitoring Plate ETX ID	Microbial ID	Notes
Personnel EM Bracketing								
Personnel (Left Touch and Right Touch)	Daily	09Oct26	CCD	Date Before Aliquoting	No Growth	Not Applicable	None	None
Personnel (Left Touch and Right Touch)	Daily	12Oct26	CCD	Date of Aliquoting	No Growth	Not Applicable	None	None
Personnel (Left Touch and Right Touch)	Daily	13Oct26	CCD	Date After Aliquoting	No Growth	Not Applicable	None	None
Biological Safety Cabinet EM Bracketing Biological Safety Cabinet (BSC)								
Surface Sampling of ISO 5 (4 locations)	Daily	09Oct26	ALA	Date Before Aliquoting	No Growth	Not Applicable	None	None
Surface Sampling of ISO 5 (4 locations)	Daily	12Oct26	CCD	Date of Aliquoting	No Growth	Not Applicable	None	None
Surface Sampling of ISO 5 (4 locations)	Daily	13Oct26	CCD	Date After Aliquoting	No Growth	Not Applicable	None	None
Settling Sampling of ISO 5 (2 locations)	Daily	09Oct26	ALA	Date Before Aliquoting	No Growth	Not Applicable	None	None
Settling Sampling of ISO 5 (2 locations)	Daily	12Oct26	CCD	Date of Aliquoting	No Growth	Not Applicable	None	None
Settling Sampling of ISO 5 (2 locations)	Daily	13Oct26	CCD	Date After Aliquoting	No Growth	Not Applicable	None	None
Weekly Active Air Sampling Bracketing								
Active Air Sampling of Cleanrooms	Weekly	12Oct26	SMO	Week of Testing	6 CFUs (ISO 8 114)	ETX-261021-0578	*Moraxella osloensis* *Sporisorium graminicola* *Staphylococcus haemolyticus* *Staphylococcus hominis* *Kocuria indica* *Ustilago maydis* *Kocuria rhizophila/ sp. BT304*	None
Surface Sampling of Anteroom and Cleanroom Bracketing								
Surface Sampling of Cleanrooms	Weekly	12Oct26	SMO	Week of Testing	No Growth	Not Applicable	None	None

*Observation is based on CFU counts